

Profile & Field of Specialization

Experimental photonics researcher and educator focused on building and validating fiber-optic and plasmonic sensing systems at the device and system levels. My work integrates fiber sensors (FBGs, DAS, and nanomaterial-functionalized microresonators) with machine-learning-assisted interrogation, emphasizing robustness, scalability, and deployability across energy, biomedical, and environmental domains.

Education

Ph.D. in Electrical Engineering (Photonics), University of Malaya, Malaysia, 2023

M.Sc. in Telecommunication Engineering, University of Malaya, Malaysia, 2019

B.Sc. in Electrical Engineering (Telecommunication & Electronics), Sana'a University, Yemen, 2016

Professional Experience

Postdoctoral Fellow, King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia, 2025–Present.

- Led the design and experimental validation of fiber-optic sensing systems, including FBG and distributed acoustic sensing (DAS) platforms, with integrated machine-learning-based signal interpretation at the device and system levels.
- Led a successfully funded internal grant project, delivering end-to-end experimental workflows from proposal development and equipment selection to experimental execution, data analysis, and dissemination through publications and patent-related activities.
- Initiated and contributed to joint projects with KFUPM research centers, including the Center for Integrative Petroleum Research and the Center for Aviation and Space Exploration, expanding sensor application scope while maintaining alignment with core photonic sensing research.
- Supervised and mentored graduate and undergraduate researchers, supporting experimental execution and data analysis; additionally, engaged in STEM research outreach for high-school students.

APD Postdoctoral Fellow, Universitas Airlangga (Indonesia), 2024–2025

- Developed and experimentally validated Au-graphene plasmonic sensing platforms under international collaborative grants.
- Coordinated cross-institutional experimental workflows and data exchange across Malaysian–Indonesian research teams.

Graduate Research Assistant, University of Malaya, Malaysia, 2020–2022

- Authored a successful industry-funded proposal (Petronas PRSB–UiTM), aligning laboratory sensor development with applied performance requirements.
- Designed, fabricated, and experimentally characterized ZnO-, graphene-, and polymer-coated fiber sensors.

Selected First Author Publications (ORCID: [0000-0002-8151-4470](https://orcid.org/0000-0002-8151-4470))

Zain, H.A., et al., “Deep Learning for Spectral Processing in Microbottle Resonator Humidity Sensors,” IEEE Sensors Letters, 2026

Zain, H.A., et al., “A Deep Learning-Based Fusion Framework for Temperature-Aware Relative Humidity Sensing Using Dual Fiber-Optic Sensors,” IEEE Access 2026

Zain, H.A., et al., “HEC/PVDF-Coated Microsphere Humidity Sensor Using CNN-Based Barcoding,” IEEE Photonics Technology Letters, 2025.

Zain, H.A., et al., “Deep Learning-Integrated Agarose-Coated Microloop Fiber Resonator,” IEEE Sensors Journal, 2025.

Zain, H.A., et al., “Graphene Oxide-Coated Microbottle Resonators,” Optoelectronics Letters, 2024.

Zain, H.A., et al., “Review: Microbottle Resonators,” Micromachines, 2023.

Zain, H.A., et al., “Machine Learning Algorithms for SPR Bio-Detection,” Journal of Physics: Conference Series, 2022.

Zain, H.A., et al., “Graphene-Gold Kretschmann SPR for Humidity Detection,” IEEE Sensors Letters, 2022.

Zain, H.A., et al., “Graphene/PVA D-Shaped Fiber Sensor,” Sensors and Actuators A, 2021.

Primary contributions across these publications include experimental conception, sensor architecture design, nanomaterial integration, and machine-learning-assisted signal interpretation.

Funding & Projects

- **KFUPM Interdisciplinary Centers Grant (INCS2611):** Proposal authorship, experimental planning, system development, and validation.
- **KFUPM Interdisciplinary Centers Grant (INCS2505):** Proposal authorship, experimental planning, system development, and validation.
- **Petronas PRSB–UiTM Collaborative Research Project:** Proposal authorship and industry-aligned fiber-optic sensor development.
- **INTI University Seed Fund Research Project:** Proposal authorship and experimental sensor design, fabrication, and characterization.

Teaching Experience

Introduction to Research (RES 100, Undergraduate) Instructor, KFUPM, Saudi Arabia, 2025–Present.

The course covered research types and databases, reading/writing scientific papers and proposals, research methods and literature review, and research presentation and dissemination.

Huda Adnan Zain
Researcher – Experimental Photonics & Fiber-Optic Sensing
IRC-CSS, KFUPM, Dhahran, Saudi Arabia | huda.zain@kfupm.edu.sa

Digital Systems Laboratory (KIE1001, Undergraduate) Lab Instructor, University of Malaya, Malaysia, 2021-2022.

The lab introduced students to breadboard implementation, combinational logic circuits, and sequential logic circuit design/analysis.

Expertise

Sensing Systems: Fiber Bragg Gratings (FBG), Distributed Acoustic Sensing (DAS), Surface Plasmon Resonance (SPR, Kretschmann), microloop and microbottle resonators

Signal Processing: ML-assisted denoising (15% noise reduction delivered), noise analysis, signal stability optimization, real-time analysis pipelines

Software and Automation: Python (instrument control, data acquisition, automated analysis), MATLAB, GDSFactory

Simulation and Design: Ansys Lumerical, COMSOL, Zemax

Instrumentation: Laser sources (tunable, ASE), optical spectrum analyzers, optical power meters, vibration generators, temperature-controlled systems, humidity-controlled systems, fiber tapering and D-shaping equipment, surface plasmon resonance setups (He-Ne, prism-coupled), and microscopy

Materials and Fabrication: Graphene-gold interfaces, ZnO nanorods, fiber tapering, side-polished fibers, nanomaterial coatings

Languages: English (C2, Fluent), Arabic (Native), German (Basic)

Membership of Professional Bodies:

- IEEE member
- Women in Engineering IEEE member
- IEEE Photonics Society Member
- IEEE SC Optical Sensors Cluster member

Reviewer in IEEE sensors journal and Journal of Lightwave Technology. Eleven manuscripts reviewed in 2026.